

- 1 The rate of heat flow through a rod due to conduction is given by Fourier's Law:

$$\frac{Q}{t} = \frac{CA(\Delta T)}{L}$$

where A is the cross-sectional area of the material,

L is the length of the material,

ΔT is the temperature difference across the length of the material, and

C is a constant.

- (a) Determine the SI base units of C .

$$\begin{aligned} [C] &= \frac{[Q][L]}{[t][A][\Delta T]} \\ &= \frac{(\text{kg m}^2 \text{s}^{-2})(\text{m})}{(\text{s})(\text{m}^2)(\text{K})} \\ &= \text{kg m s}^{-3} \text{K}^{-1} \end{aligned}$$

[Recall : Q can be considered as energy whose unit can be found from work done]

SI base units = [2]

- (b) An experiment is conducted to determine the value of C . Using copper rod of diameter 0.80 cm but different length, and two ends of the rod are maintained at pure ice point and steam point, the rate of flow of thermal energy was measured using a heat flux sensor. A graph of how $\frac{Q}{t}$ varies with $\frac{1}{L}$ is plotted, as shown in Fig. 1.1.

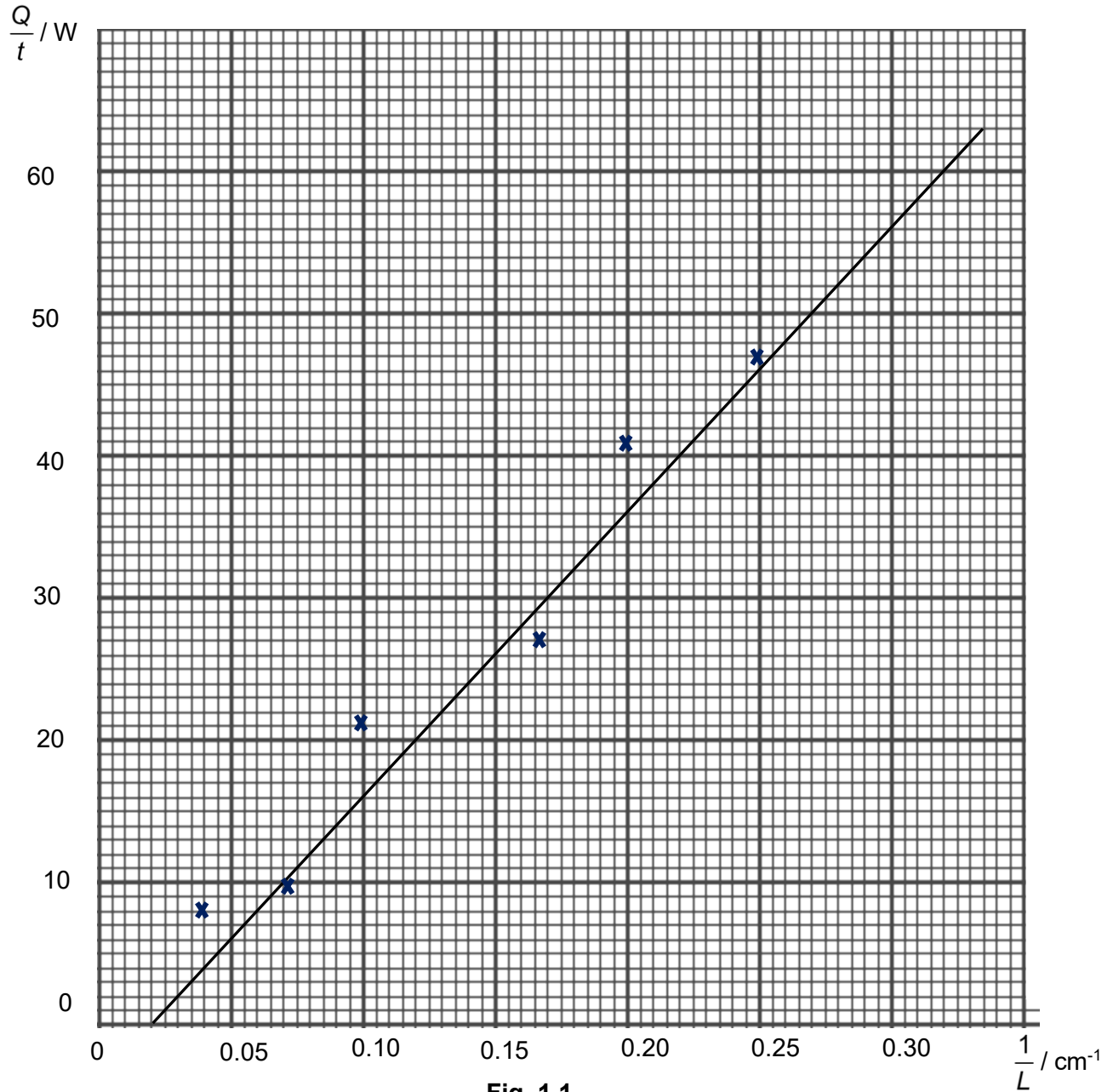


Fig. 1.1

- (i) State the feature of the graph that indicates the presence of systematic error in the experiment.

The evidence of systematic error is seen by the graph shift by a fixed value to the right and did not pass through the origin. The best fit should pass through the origin if there is no systematic error.

(ii) Calculate the value of C , in SI units, from Fig. 1.1.

gradient of graph = $CA\Delta T$

$$\text{Gradient} = \frac{56.0 - 10.0}{0.300 - 0.070}$$

$$= \frac{46.0}{0.230}$$

$$= 200 \text{ W cm}$$

$$= 2.00 \text{ W m}$$

$$C = \frac{\text{gradient}}{A\Delta T}$$

$$= \frac{2.00}{\left(\pi \left(\frac{0.80 \times 10^{-2}}{2} \right)^2 \right) (100 - 0)}$$

$$= 398 \text{ kg m s}^{-3} \text{ K}^{-1}$$

value of C = SI base units [3]

(iii) With reference to Fig. 1.1 and (b)(i), state and explain whether the accuracy of C is affected by the presence of systematic error in the experiment.

No, the gradient of the straight line is equal to $CA\Delta T$ and the gradient is not affected by the presence of systematic error, the accuracy of C is not affected.

.....
[1]

[Total : 7]

